

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Engineering Chemistry [CHM-1071]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) i) Explain the construction and discharging reactions of lithium ion batteries. Write any two advantages of it. (4)
- A) ii) Discuss the influence of the pH and current density on electro deposition.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) i) What is the pH of the solution when the cell $\text{Pt} | \text{H}_2 | \text{H}^+_{(x)} || \text{KCl}_{(0.5 \text{ M})} | \text{Hg}_2\text{Cl}_2 | \text{Hg}$ is set up at 298 K, which provides an EMF of 1.11 V? (EMF of normal calomel electrode is 0.2810 V). (2 Marks)
- ii) Describe the construction and working of alkaline fuel cell (2 Marks)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Write any two differences between: (2)
- i. Galvanic cell and fuel cell
- ii. Electroplating and electroless plating

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) The concentration of yeast t-RNA in an aqueous solution is 10 M. The absorbance is found to be 0.209 when this solution is placed in a 1.00 cm cuvette and 258 nm radiations are passed through it. (2 Marks) (4)
- A) a) Calculate the specific absorptivity
- b) What will be the absorbance if the solution is 5 M?

ii) With relevant graphical representation explain the conductometric titration of weak acid vs strong base. (2 Marks)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Define reverse osmosis. Describe with a neat diagram how sea water is purified using this technique. Write any two advantages of it. (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) i) Define electrodialysis. (2)
- ii) Distinguish between hard water and soft water

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3) i) How galvanic series is different from electrochemical series? (1 Marks) (4)
 ii) Explain the electrochemical theory of corrosion. (3 Marks)

A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) i) How do the following factors affect the rate of corrosion? (2 Marks) (4)

- pH of the medium.
- Relative cathodic and anodic area.

- ii) Discuss the importance of selection of proper material and good design for corrosion control. (2 Marks)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- C) Identify and explain the following forms of corrosion: (2)

- i) Copper piping connected to steel tanks.
 ii) Deposition of extraneous matter on the surface of iron.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) i) Differentiate between the following (Any two points each) - (2 Marks) (4)

A)

- a) Thermotropic and lyotropic liquid crystal.
 b) Matrix and dispersed phase.

- ii) Explain the chemical vapour deposition process of thin film fabrication with suitable example. (2 Marks)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- B) i. Polyethylene sample has the following composition (3 Marks) (4)

Degree of polymerization	400	300	200	100
% of composition	10	20	30	40

Calculate the number average molecular weight, weight average molecular weight and PDI of the given polymer sample

- ii. Why polyvinyl chloride (PVC) has higher strength than Polyethylene? (1 Mark)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain the sol-gel method for the preparation of nanomaterials. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5) i) Define decomposition potential of an electrolyte. (1 Mark) (4)

- ii) At 80 °C, temperature co-efficient = -1.75×10^4 V/K and $E_{\text{cell}} = 2.89$ V for the cell $\text{Pt} | \text{H}_2(\text{g}) ||$

- A) $\text{HCl}(\text{aq}) | \text{Cl}_2 | \text{Pt}$. Calculate the free energy, enthalpy and entropy for the cell reaction - $\text{H}_2(\text{g}) + \text{Cl}_{2(\text{g})} = 2\text{HCl}(\text{aq})$ (3 Marks)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) i) Explain the preparation of nanomaterials by ball milling method. (2 Marks) (4)

- ii) Explain the working of liquid crystal displays in calculators. (2 Marks)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the role of addition of Hydrazine (N_2H_4) and ZnSO_4 in corrosion inhibition (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CHM-1071:Engineering Chemistry

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	4
3	3	CO 3	12
4	4	CO 4	6
5	5	CO 5	14
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CHM-1071:Engineering Chemistry

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	11	34(68%)
3	Application	4	16(32%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)